

Health, Safety and Wellbeing (HSW) Risk Assessment (Department of Electrical, Computer, and Software Engineering)

Project Activity Name:	Part 4 Project #43: Reconfigurable Neural Processing Unit (NPU) for Energy-Efficient AI at the Edge		Assessment Date:	04/04/2025	
Student/Visitor/Consultant Name:	Kristelle Sampang		UoA ID & UPI:	463680577/ksam836	
Program:	☒ UG ☐ Intern ☐ MES ☐ ME ☐ PhD ☐ PostDoc ☐ Consultant ☐ Visitor				
Supervisor (Responsible Line Manager):	Morteza Biglari-Abhari		Activity Location:	Embedded Systems Lab	
Risk Rating of Highest Risk Task:	☒ Low (1-2) ☒ Moderate (3-4) ☐ High (6-9) ☐ Extreme (12-16)				
Brief Project Activity Description:	Developing a testbench setup that can be used to test and evaluate approach of synchronous programming language for autonomous vehicles.				
By completing and signing this document both parties, the student/visitor/consultant and the project supervisor (i.e. the responsible line manager), accepts the following terms – Read the UoA 'Health and Safety' policy document – All foreseeable risks, risk ratings and controls associated with the project activities have been identified – Any project activity must take place within the laboratory space allocated by the technical staff – Technical staff member in-charge of the space has been briefed of the risks and control measures – This document is revised and updated if/when the project activities or space allocation or equipment usage change – Any incidents are reported and appropriate action taken – Student/visitor/consultant has competently completed the general/power lab safety induction and any other additional training required – Student/visitor/consultant is not allowed to be engaged in project activities other than what is captured in this document – It is the project supervisors (i.e. the line managers) responsibility to ensure the control measures are effectively implemented and they are reviewed if/when needed					
Signed by the supervisor (i.e. responsible line manager):			Signed by the student:		
Morteza Biglari-Abhari <i>Morteza Biglari-Abhari</i> 04 / 04 / 2025			Kristelle Sampang Kristelle 04/04/2025		
Name Signature Date			Name Signature Date		

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Identify Hazards and Control the Risks:

1. An activity may be divided into tasks. For each task identify the hazards and associated risks. Also list the possible scenarios which could sooner or later cause harm.
2. Determine controls necessary based on University standards, legislation, codes of practice, AS / NZ standards, manufacturer's instructions etc.
3. List existing risk controls (take credit for what you do)
4. Rate the risk once all controls are in place using the matrix in
5. List any additional controls that need to be implemented and take action
6. Communicate the findings

The boxes will resize to suit your situation/the amount of text you need to use – press tab after last cell to create new rows

Task	Associated Hazard	Who may be harmed	Existing controls	Current Risk Rating (L)Likelihood x (C)Consequence = (R)Rating			Additional Controls
				L	C	R	

Develop control circuitry utilizing high voltage	*Electrical Shock *Short Circuit	Students, Staff, Visitors and Equipment	Isolating transformer, insulation	1	1	1	* You are not allowed to touch any equipment, project or tools that do not belong to you. *You have to have an official safety induction before entering any lab or working on any project at the university.
Mains cable laid across lab floor Trip hazards from general debris	*Trip hazard	Students, Staff, Visitors and Equipment	Use adequate indication to warn others about the hazard and use cable duct	2	1	2	*Mind your steps and be cautious /aware of your surroundings. * Take care, be tidy and keep a good practice of housekeeping. *Tell the lab manager.

(Please specify based on your lab activity)							
Use un protected circuit	*Burn *Short circuit * Electrical Shock	Students, Staff and Visitors	Use portable RCD Ask your supervisor to check your connection Have at least one person with you in the lab during your operation of the circuit	2	1	2	*You have to have a safety induction before entering any lab or working on any project at the university.
Deal with wet-cell batteries	*Burn *Foreign objects go into the eyes *Electrical Shock *Short Circuit	Students, Staff, Visitors and Equipment	Safety glasses Gloves	1	2	2	*You have to have a safety induction before entering any lab or working on any project at the university.
Using instrument and electrical equipment while the PAT test is out of date	*Electrical Shock *Short Circuit	Students, Staff, Visitors and Equipment	PAT testing, TUFTAG date	1	1	1	*You have to have a safety induction before entering any lab or working on any project at the university. *Check the PAT test tag is on the equipment that you are about to work on.
Any cause of personal injury, (Please specify	*Small cut	Students, Staff and Visitors	First Aider with First Aid kit	2	1	2	*You have to have a safety induction before

based on your lab activity)	*Small injuries that do not need medical attention						entering any lab or working on any project at the university. * Take care, be tidy and follow the practice of a good housekeeping. *Mind your steps and be cautious /aware of your surroundings.
Any cause of fire (Please specify based on your lab activity)	*Small or serious injuries.	Students, Staff, Visitors and Equipment	Fire alarm and Fire Warden	2	1	2	*Raise the fire alarm. *You have to have a safety induction before entering any lab or working on any project at the university. * Take care, be tidy and keep a good practice of housekeeping. *Mind your steps and be cautious /aware of your surroundings.
Overheated circuits (Please specify based on your lab activity)	*Electrical Shock *Short Circuit *Burn	Students, Staff, Visitors and Equipment	Use current limit	2	1	2	*You have to have a safety induction before entering any lab or working on any project at the university.

							* Take care, be tidy and keep a good practice of housekeeping. *Mind your steps and be cautious /aware of your surroundings.
Use of Power supply (Please specify based on your lab activity)	*Electrical Shock *Electric Fault *Overheated circuits	Students, Staff, Visitors and Equipment	RCD protected No liquids or interference around your work stations Use current limit	2	1	2	All mains protected with RCD in the lab You have attended compulsory equipment induction showing how to use equipment safely in the lab or working on any project at the university. No food and drink allowed in the lab.
Use of Oscilloscope	*Electrical Shock *Damaging Scope *Short circuit	Students, Staff, Visitors and Equipment	Use of isolation transformer if measuring mains (AC voltage)	2	1	2	Follow Lab induction manual and operating instructions of oscilloscope specifying safe use of oscilloscopes
Use of soldering irons	*Burn *Harmful fumes	Students, Staff, Visitors and Equipment	Use safety goggles, closed shoes, extraction fan, wear long sleeve	2	1	2	Lab inductions specifying the

							importance of lab safety around soldering irons
Computer works	*Back and neck pain *Tired Eyes *Headache	Students, Staff and Visitors	Sit up straight Take regular breaks Use medical glasses if needed Drink enough water	2	1	2	Read the practical guide to carry out a Workstation Assessment available on The University of Auckland website

Action Plan						
Management agreed additional control measures to be implemented	Resources Required	Action By:			Action Complete: Responsible Line Manager	
		Responsible Person	Target Date	Completion Date	Signature	Date

Review	
Review Details	Comments
Scheduled Review Date	
Are all control measures in place?	
Are controls eliminating or minimising the risk?	
Are there any new problems with the risk?	
Are the supervisory arrangements adequate?	
Are the levels of skills, capabilities and training adequate?	
Review By: (name)	
Review Date:	

Communication				
	Method	Yes	Date	Comments
Reference of formal communication to staff	Copy of risk assessment issued to staff			
	Controls covered in team procedure issued to staff			
	Staff handbook issued to staff			
	Other			
How they were consulted on the risk	Health, Safety and Wellbeing Committees			
Additional Methods of Communication	Induction			
	Toolbox Talk			
	Team Meeting			
	Email circulation			
	Other			

HSW Risk Assessment Matrix						
Likelihood Level	4	Very likely Probably expect the event to occur in most circumstances	Moderate (4)	High (8)	Extreme (12)	Extreme (16)
	3	Likely Event likely to occur at least once over the coming year	Moderate (3)	High (6)	High (9)	Extreme (12)
	2	Possible Event may occur at some time	Low (2)	Moderate (4)	High (6)	High (8)
	1	Unlikely Occurrence is conceivable, but not expected to occur	Low (1)	Low (2)	Moderate (3)	Moderate (4)
			Minor	Moderate	Major	Severe
			1	2	3	4
			Consequence level			
Consequence description	Harm to People Potential for injury or death		None or trivial / negligible injury (no or slight injury which requires localised first aid)	Minor injury (illness or injury is not serious, medical treatment required)	Serious injury (serious injury or illness, hospitalisation required)	Fatality, major injury (death, permanent disablement, or significant long-term illness)
	People Affected Extent of people potentially affected		None or few (e.g. 0 to 2)	Small numbers (e.g. 3 to 10)	Moderate numbers (e.g. 10 to 50)	Wide scale (e.g. more than 50)
	Reputation and Legal Potential for publicity with a negative impact on reputation / potential for legal prosecution		None or issue raised by staff or students and resolved promptly by management None or legal dispute – found not guilty – fines up to \$3 million (Body Corporate), \$600,000 (Officer)	Internal scrutiny to prevent escalation and short-term stakeholder concern Minor non-compliance, limited notification to regulators / affected stakeholders	Medium-term stakeholder concern, national media scrutiny and 'brand' impact Medium non-compliance, moderate notification to regulators / affected stakeholder, potential for legal proceedings / fines	Persistent stakeholder concerns, international media scrutiny and long term 'brand' impact Significant non-compliance, extensive notification to regulators / affected stakeholders, potential for legal proceedings / imprisonment / fines

	Operations Extent of ability to maintain core business	None or business interruption < 4 hours None or effectiveness and efficiency of a service, programme or project impacted in the short term None or slight damage to property or equipment	Business interruption between 4 hours to 5 days Operational disruption manageable by workarounds Moderate damage to property or equipment	Business interruption > 5 days Medium operational impact resulting in delay of key deliverables Major damage to property or equipment	Business interruption of many weeks Breakdown of key activities and significant long-term impact Massive damage to property or equipment
	Environment Extent of negative impacts on the environment	None or minimal impact None or clean up expenses up to \$25,000	Minor short-term or intermittent impact, able to be contained with specialist assistance Clean up expenses up between \$25,000 to \$1m	Serious, medium-term detrimental impact Clean up expenses up between \$1m - \$5m	Very serious, long-term or permanent damage Clean up expenses > \$5m

Consider the Likelihood

How often is the task done? Has an accident happened before (here or at another workplace)? How long are people exposed? How effective are the control measures? Does the environment affect it (e.g. light, temperature, space)? What are people's behaviours (e.g. stress, panic, deadlines)? What people are exposed (e.g. disabled, young students, etc.)?

Consider the Consequences

What type of harm could occur (minor, serious, death)? Is there anything that will influence the severity (e.g. proximity to hazard, person involved in task, etc.)? How many people are exposed to the hazard? Could one failure lead to other failures? Could a small event escalate?

Calculate the Risk

The final score for each risk is calculated by multiplying the likelihood and consequences response scores. This will give a risk score of between 1 and 16.

All risks rates as "High" or "Extreme" require detailed analysis of mitigating practices / controls to determine the residual risk rating. **Action must be taken.**

"Low" and "Moderate" risks may be excluded from further analysis (other than when the consequence may be severe). However the rationale for excluding these risks should be documented to demonstrate the completeness of analysis undertaken. **Some action may be required.**

Other than in the most unlikely circumstance, risks that can cause major or severe harm to people have been determined as "high" or "extreme". Management review is considered appropriate for risks of these nature due to the potential magnitude of the impact, even though the likelihood may be assessed as relatively low.

Risk Priority - Legend

Extreme (12-16)	Intolerable risk. Immediate action(s) is to be taken by Faculty/Service HSW risk owners - including DVCs, Deans of Faculties, Directors of Services, Academic Heads/PIs, Services Managers. Work should not be started or continued until the risk has been reduced to as low as reasonably practicable using the hierarchy of risk controls. The Associate Director Health, Safety and Wellbeing, and Manager Risk and Performance must be advised of the risk for their review. The risk should be included in the UoA wide risk register.
High (6-9)	Should not be tolerated. Urgent action is to be taken by the immediate manager. Work should not be started or continued until the risk has been reduced to as low as reasonably practicable using the hierarchy of risk controls. The HSW Manager working with the Faculty/Service, and Manager Risk and Performance must be advised of the risk for their review. To be included in the UoA wide risk register.
Moderate (3-4)	Management to monitor risks in case changing circumstances increase the level of risk. Some action may be required, e.g. improving controls.
Low (1-2)	Requires no further attention above routine practices and procedures, apart from monitoring.

Note: This proposed Health and Safety Risk Assessment Matrix aligns with WorkSafe NZ guidance, UoA Resilience Management Plan, UoA Risk Determination Matrix, UoA TVRA and UoA Incident Levels